

- Q.23 Explain effect of noise on FM carrier.
- Q.24 What is need of pre-emphasis and de-emphasis?
- Q.25 Explain working operation of balanced modulation.
- Q.26 Write short note on PLL.
- Q.27 Differentiate between narrowband and wideband FM.
- Q.28 State advantage, disadvantages and applications of SSBSC modulation.
- Q.29 What is the concept of AFC?
- Q.30 Explain the sampling Theorem.
- Q.31 Compare Modulation Demodulation.
- Q.32 Explain phase modulation in brief.
- Q.33 What is Modulation brief.
- Q.34 What is the function of transmitter in the communication system.
- Q.35 Compare the PAM and PWM.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain principle of demodulation of AM wave using diode detector circuit.
- Q.37 What are the basic concepts of TDM and FDM, explain in detail.
- Q.38 Write a short note on following:
- Explain the Carson's rule
 - Explain Square law modulation.

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Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 In amplitude modulation, which among the following is constant?
- Amplitude
 - Frequency
 - Wave length
 - Time period
- Q.2 Baseband compression produces_____
- a small range of frequencies from low to high
 - a small range of different phases
 - a small range of angles
 - a small range of amplitude
- Q.3 Frequency division multiplexing : Frequency slots: time division multiplexing:
- Time slots
 - Coded Information
 - Pulsed Information
 - Band slots
- Q.4 Which of the following receiver does not have an amplitude limiter stage?
- AM
 - PM
 - FM
 - Both AM and FM

- Q.5 What is the modulation used to?
- Reduce the bandwidth
 - Separate the transmissions of different users
 - Ensure that intelligence may be transmitted to long distances
 - Allow the use of practical antennas
- Q.6 In a PWM inverter, if the frequency of the lowest harmonics is 180Hz, then the frequency of the fundamental component would be _____
- 50 Hz
 - 60 Hz
 - 540 Hz
 - 90 Hz
- Q.7 FDM stands for _____
- Frequency Density Multiplexing
 - Frequency Difference Multiplexing
 - Frequency Division Multiplexing
 - Frequency Data Manager
- Q.8 Which of the following device was used in early TDM/PAM telemetry systems?
- Commuter
 - Linear Switch
 - Logic gates
 - DSP
- Q.9 In AM receiver, the oscillator frequency is always _____
- lower than signal frequency
 - higher than signal frequency
 - equal to signal frequency
 - equal to 1KHz

- Q.10 What is the full form of AFC?
- Amplitude to frequency conversion
 - Automatic frequency conversion
 - Automatic frequency control
 - Audio frequency control

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 FDM technique is based on the _____
- Q.12 Greater the redundancy lesser is the bandwidth expansion. (True/False)
- Q.13 Write full form of VSB.
- Q.14 QAM is a combination of _____
- Q.15 What is the line connection the positive and negative peaks of the carrier waveform called?
- Q.16 The ratio between the modulating signal voltage and the carrier voltage is called?
- Q.17 What is the reference line for the modulating signal?
- Q.18 ISB stands for _____
- Q.19 Single sideband Modulation (SSB) is generally reserved for point-to-point communication. (Yes/Not)
- Q.20 PLL means _____

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Why need of modulation?
- Q.22 Write comparison of AM and FM in communication system.